Data Science Association

Exploring NYC Rent Stabilization

Agenda

1. Definition of Rent Stabilization
2. Motivation
3. Data Description & Methods
4. Model Results & Findings
5. Conclusions

Meet The Team

Member 1

Major(s)

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Member 2

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Member 3

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Project Manager

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Definition of Rent Stabilization

Image Generated by Google Gemini
Image: Drone View of Bronx

What is Rent Stabilization?

Rent stabilization is a system **limiting** how much landlords can **raise** rent. It ensures rent can only increase by a set percentage each year and is designed to keep housing **affordable.**

Around **~1 million** apartments in NYC are rent stabilized

Recent Policies in NYC

Limits (Set annually by NYC Rent Guidelines Board):

- **2024–2025:**
  - ~2.75% (1-year lease)
  - ~5.25% (2-year lease)
- **2025–2026:**
  - ~3% (1-year)
  - ~4.5% (2-year)

Housing Stability & Tenant Protection Act (HSTPA) - 2019

- Limited how much landlords can raise rent after renovations
- Stronger protections against eviction
- One of the biggest expansions of tenant rights

Good Cause Eviction Law - 2024

- Protects more tenants (even some not stabilized)
- Limits when landlords can evict or raise rent too much

Motivation

Image Generated by Google Gemini
Image: Drone View of Brooklyn

The Debate

Inspired by **Zohran Mamdani** recent campaign to freeze rent we want to see what properties about New York City affect Rent.

We utilized **regression models** on a dataset to see which variables **contribute** to **increased** rent prices and **rent burden**.

Data Description & Methods

Image Generated by Google Gemini
Image: Drone View of Manhattan

The Data

- **Dependent Variable:**
  Median Rent

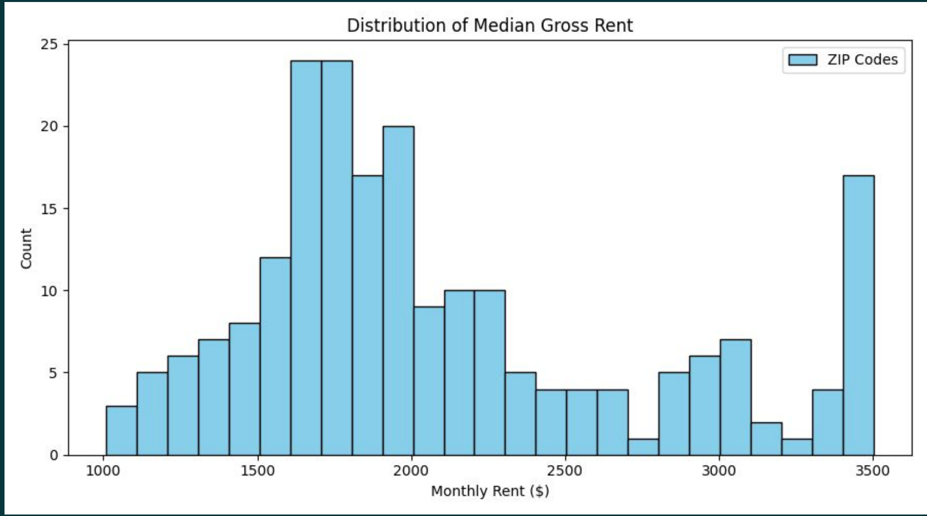
- **Socioeconomic Controls:**
  Median Household Income, Population

- **Housing Structure Variables:**
  Total Residential Units, Housing Age (Year Built), Rent Stabilization Share

- **Spatial Identifiers:**
  ZIP Code, Borough

Major Data Sources

- U.S. Census Bureau (ACS + TIGER/Line Shapefiles)
- NYC Rent Guidelines Board Stabilized Building Lists
- NYC MapPLUTO Property Database



Why Lasso Regression?

- Handles **multicollinearity**
- Performs feature selection
- Identifies **strongest** predictors while **removing insignificant** predictors

Why Multiple Linear Regression?

- **Identifies** relationship between **variables** and **outcome**
- **Easily** Interpretable
- Can be used on **numerical** and **categorical** variables

Outcome variable: **Median rent**

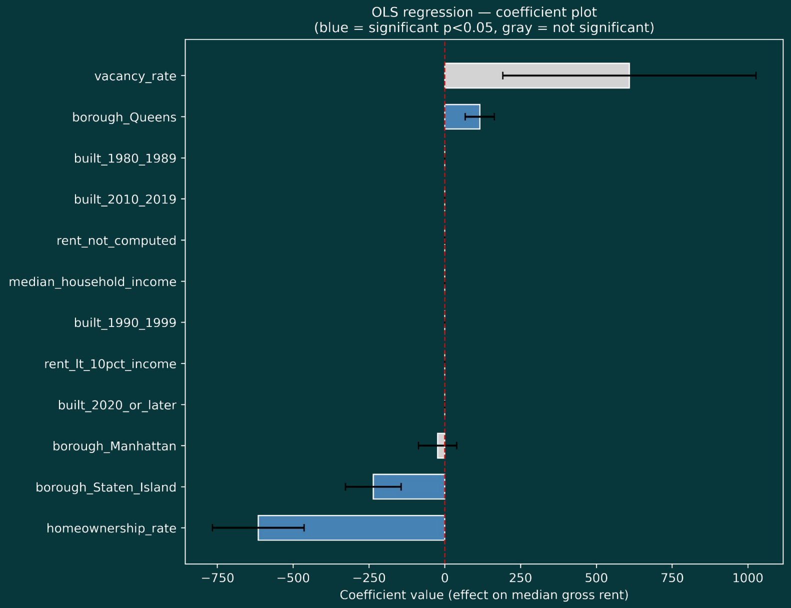
Model Results & Findings

Image Generated by Google Gemini
Image: Drone View of Queens

Income, building age, borough, and homeownership remain important predictors.

Lasso regression, after deleting extreme predictors through *Cook's Distance*.

Several variables (e.g. vacancy rate) did not reach statistical significance.



Baseline Borough: Bronx

	Test R^2	RMSE
Result	0.871	$238.95

Stabilization share adds little explanatory power in the final model.

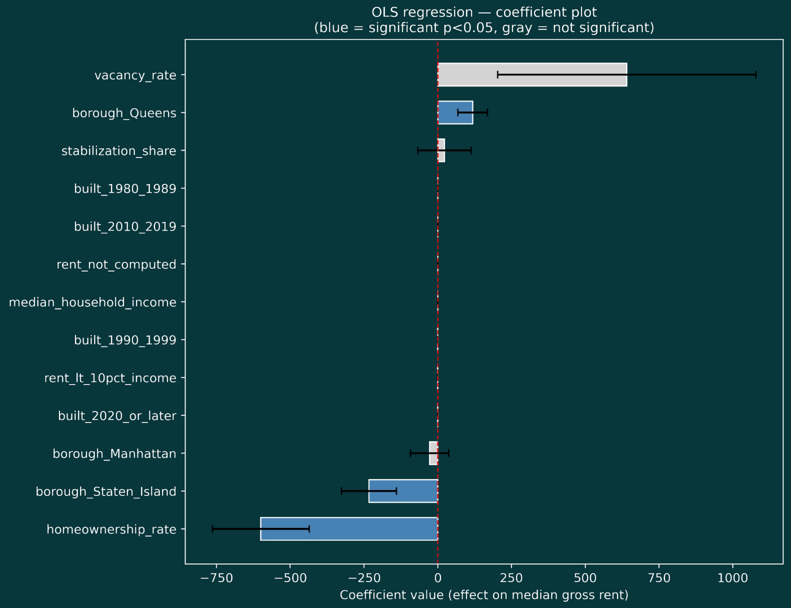
Added *stabilization share* variable from the previous model.

Intensity variable (stabilization share) is not statistically significant.

→ Suggests other variables (e.g. Year of Construction, boroughs) matters more.

Baseline Borough: Bronx

	Test R^2	RMSE
Result	0.87	$244.62



Conclusions

Image Generated by Google Gemini
Image: Drone View of Staten Island

Stabilization policy is not associated in influencing the rent price in New York City.

Controversial arguments

- NJ and CA showed slower housing stock growth compared to cities without laws.
- May help current tenants in the short-run, but potentially decreases affordability.

Foreign Countries that tried similar policies:

- Vienna, Austria: social housing without no quality degradation
- Buenos Aires, Argentina: rent increase rate (62%) > inflation rate (36%)

Limitations:

- Linear regressions confined the models to be in various relationship.
- More longer time period will help the analysis.

https://tinyurl.com/ctrl-rent

Sources

Rent Stabilization

US Supreme Court rebuffs challenge to New York rent stabilization | Reuters

Rent Increases · NYC311

2025-26 Apartment/Loft Order #57